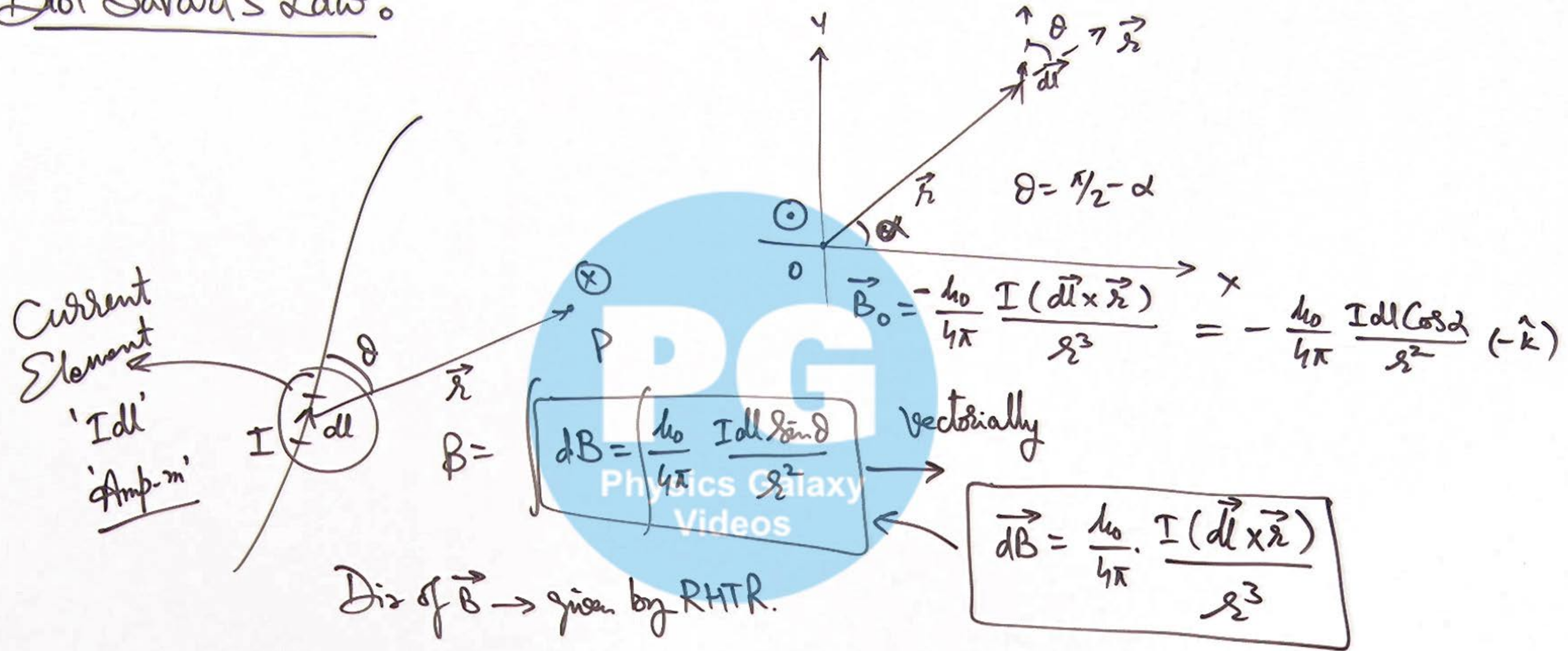


① Magnetic Field & Magnetic Forces:

- # Static charges $\longrightarrow$ produces only EF & only EF can exert force on these.
- # Moving charges $\longleftrightarrow$ produces both EF & MF & Both fields can exert force on these
- # Current Carrying Conductors (uncharged) $\longrightarrow$ produces only MF & only MF can exert force on moving charges

② Biot Savart's Law:



② Magnetic Field due to Moving Charges: (BSL)

Equivalent instantaneous
current element of a
moving charge

$$Idl \equiv qv$$



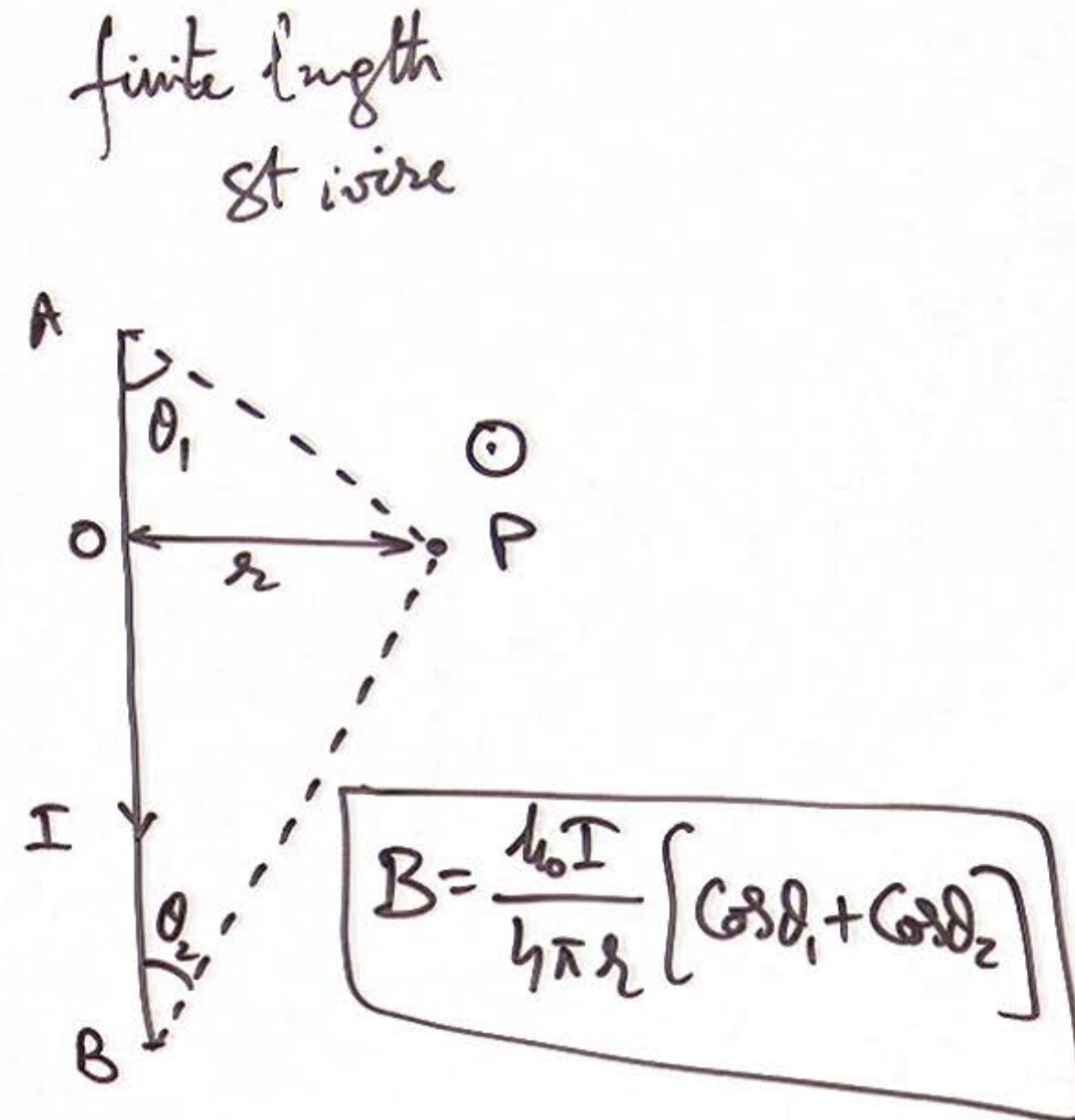
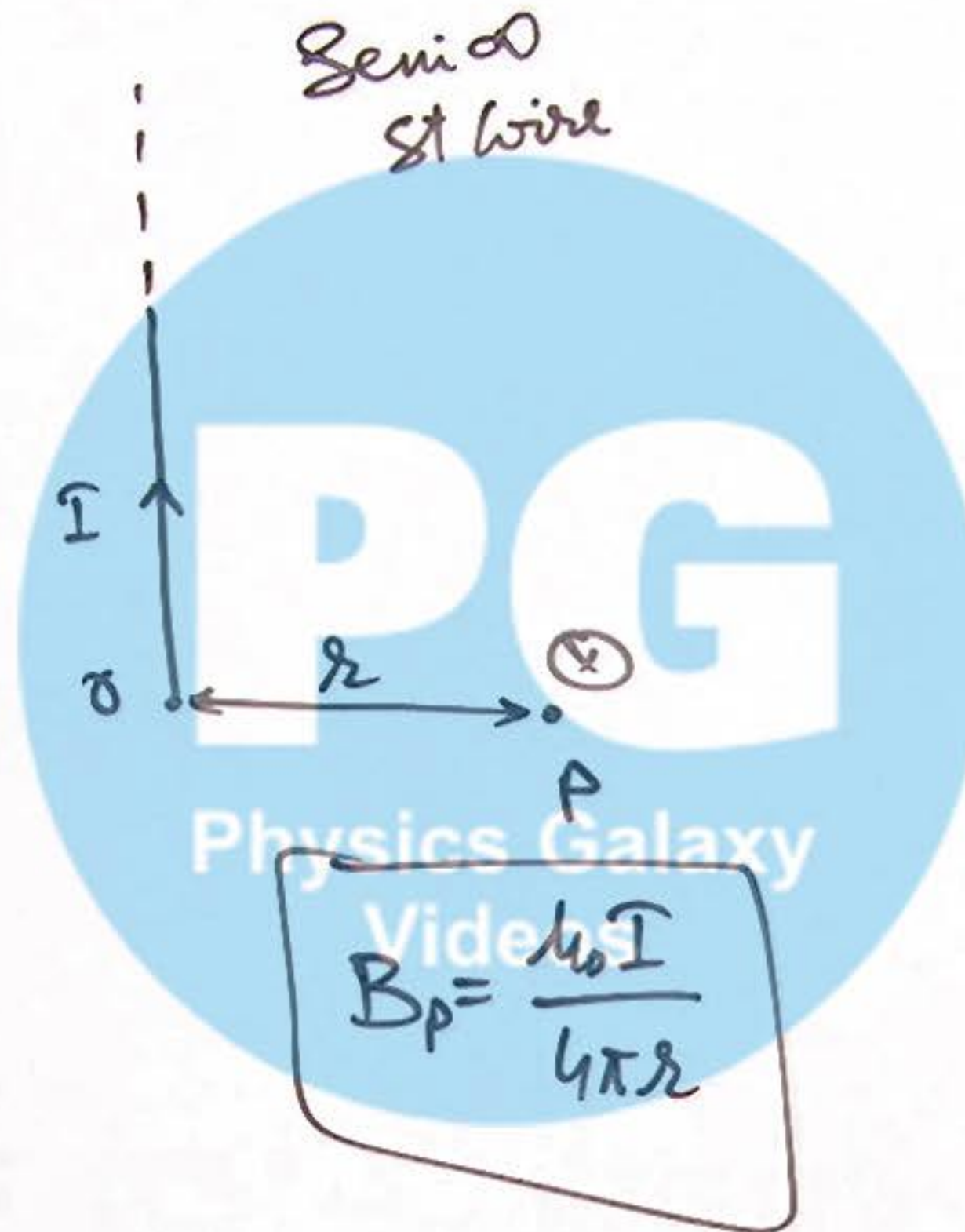
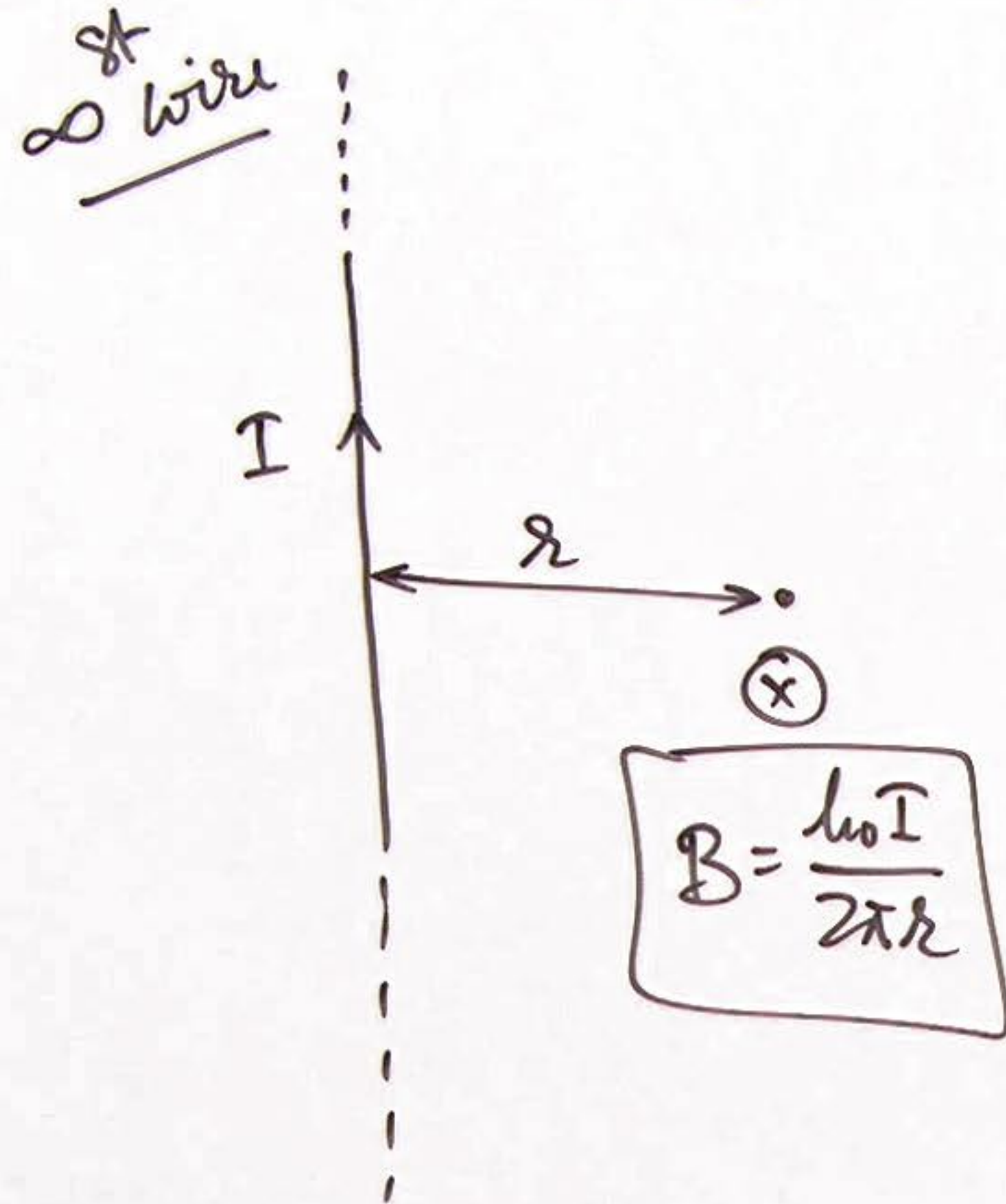
⊗ (Idl)

$$B = \frac{\mu_0}{4\pi} \frac{qv \sin \theta}{r^2}$$

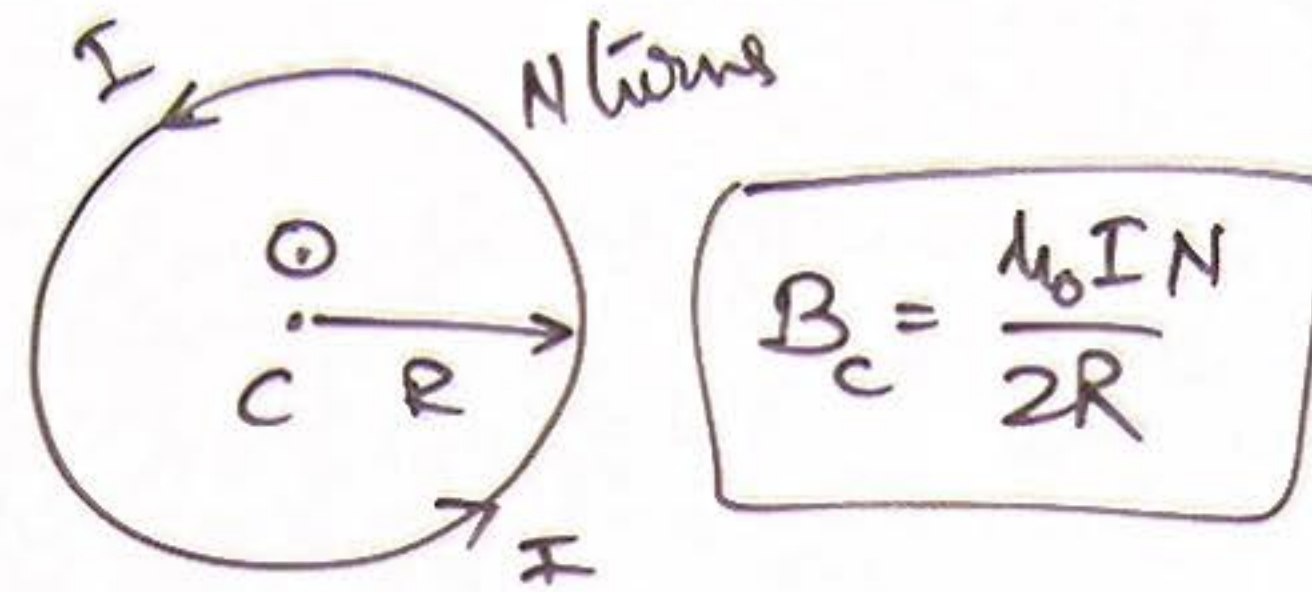
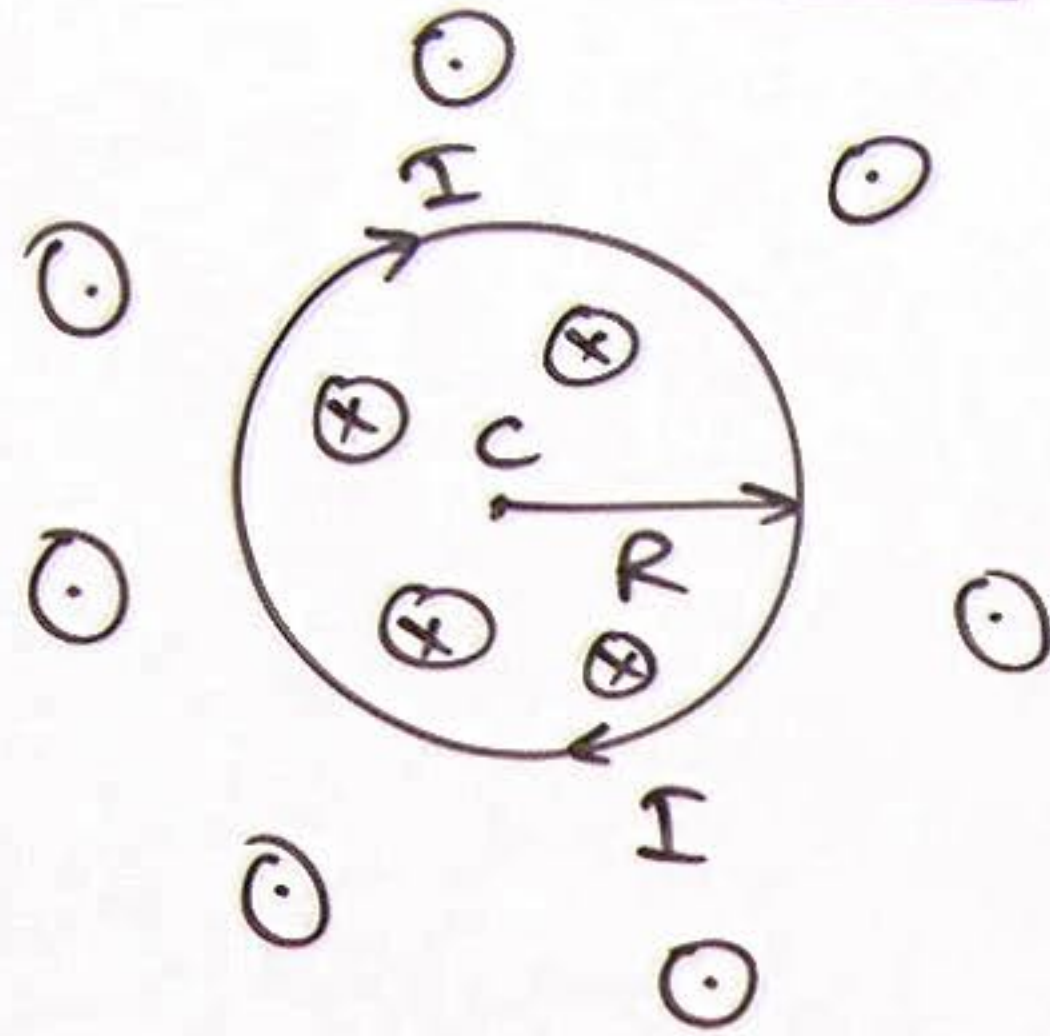
Vectorially

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{q(\vec{v} \times \vec{r})}{r^3}$$

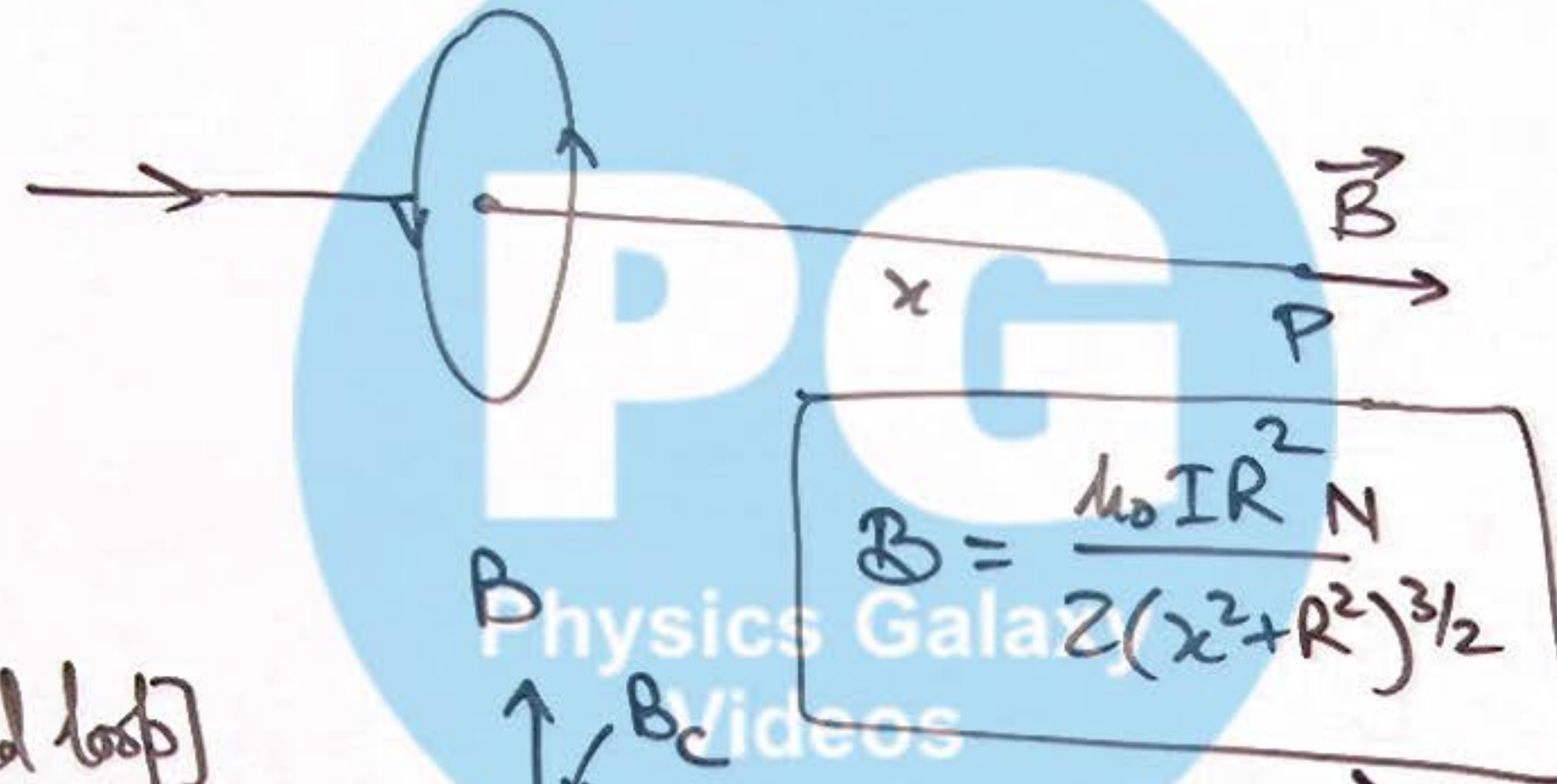
④ Magnetic Field due to a Current Carrying wire:



5 Magnetic Field due to a Circular Coil :



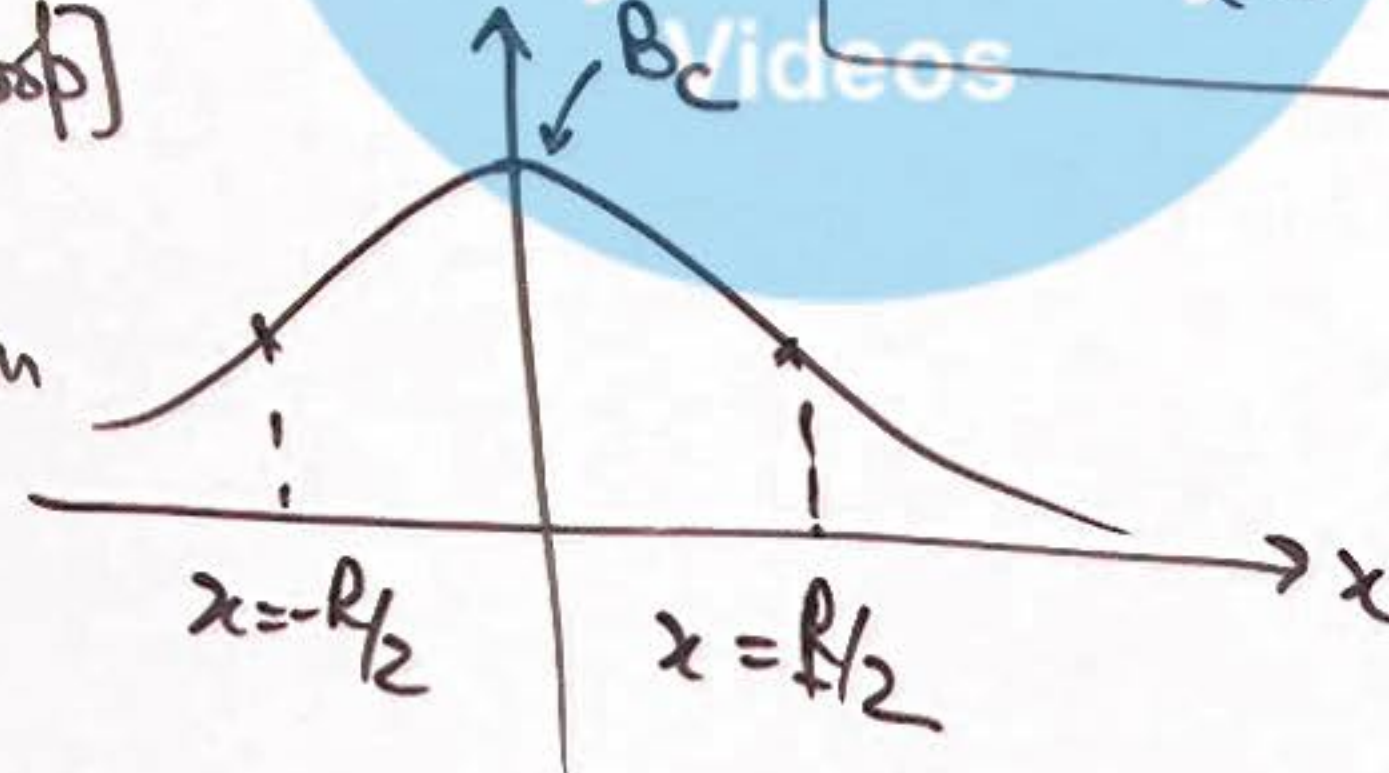
$$B_c = \frac{\mu_0 I N}{2R}$$



$$B = \frac{\mu_0 I R^2 N}{2(x^2 + R^2)^{3/2}}$$

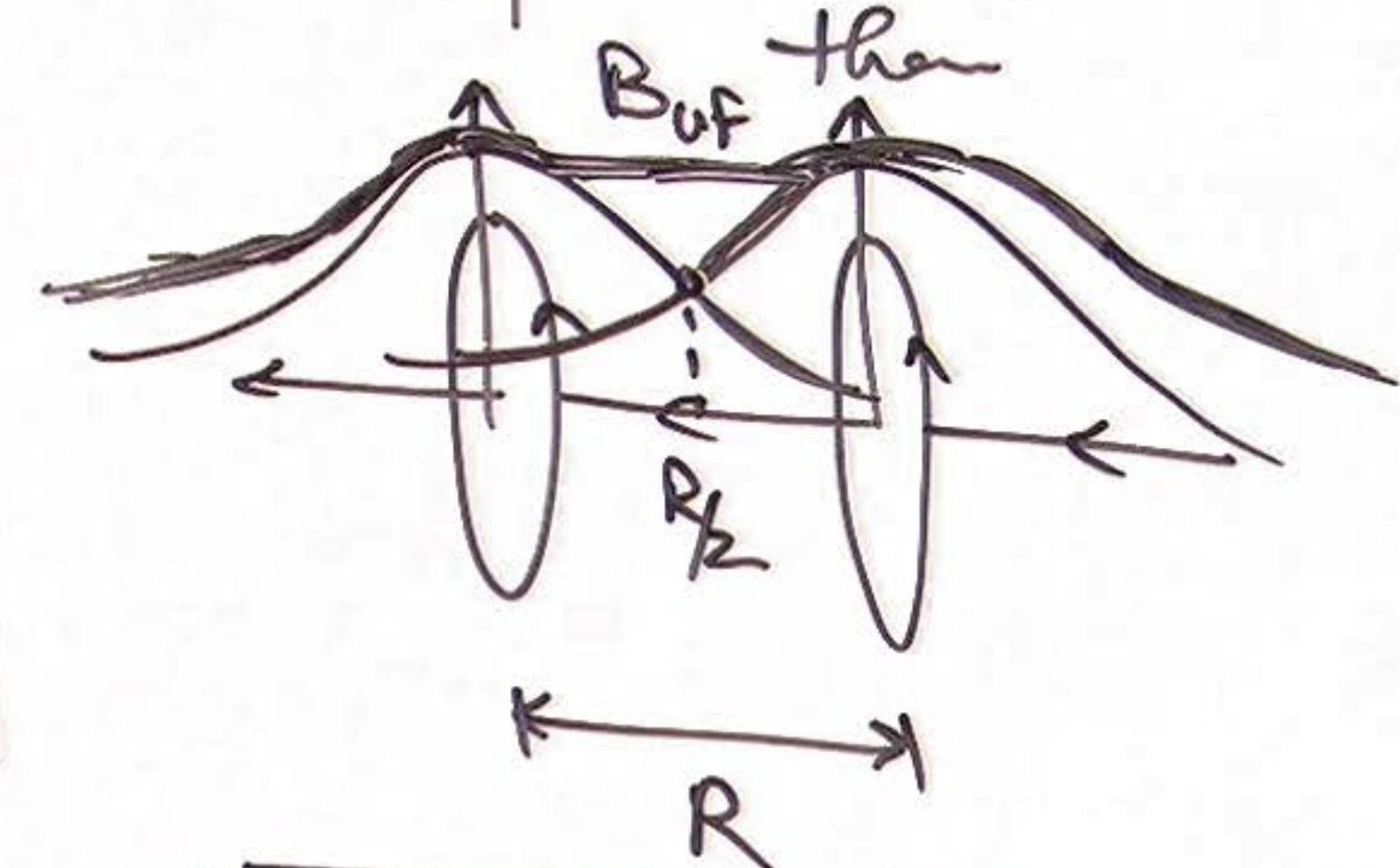
For any shape 'coil' [closed loop]

MF inside & outside are in opp direction



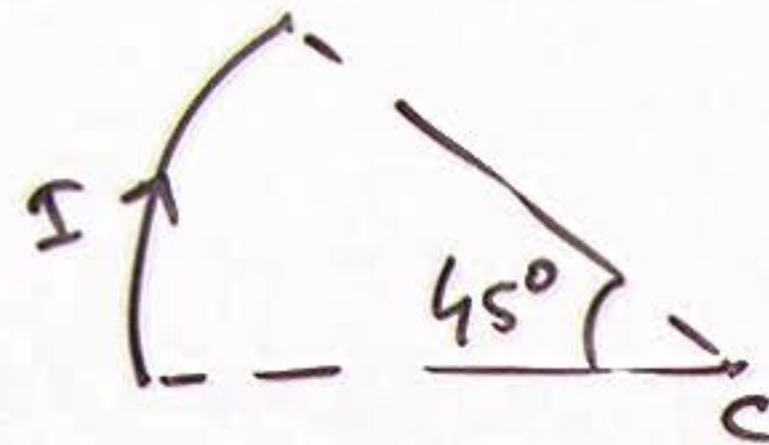
6 Helmholtz Coils :

pair of coils sep by R which produce almost uniform MF in between

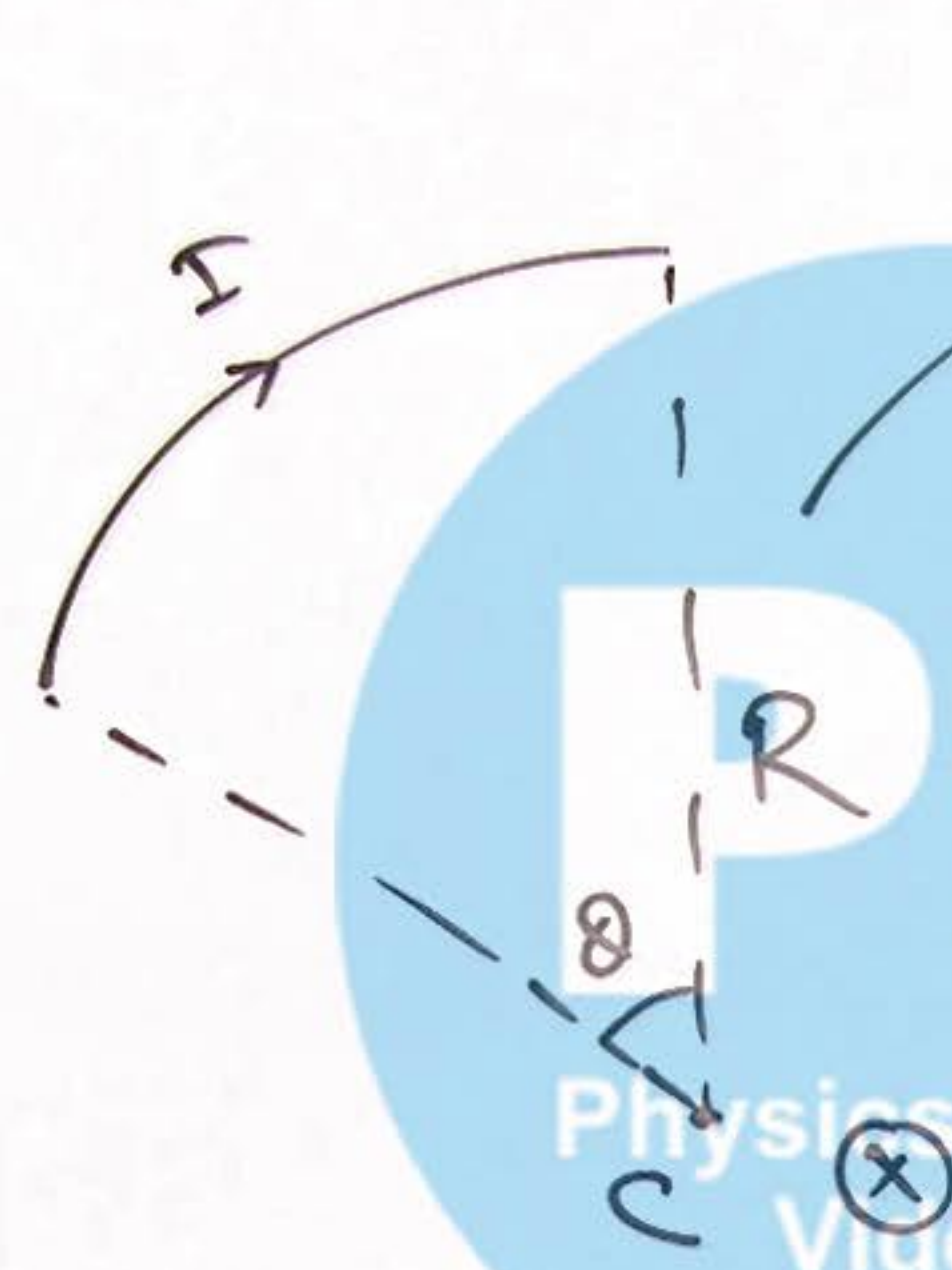


$$B_{uf} = \frac{8}{5\sqrt{5}} \cdot \frac{\mu_0 I N}{R}$$

⑦ MF at Center of a Circular Arc:



$$B_c = \frac{\mu_0 I}{16R}$$



$$B_c = \frac{\mu_0 I \theta}{4\pi R}$$

$\theta = 2\pi \rightarrow B_c = \frac{\mu_0 I}{2R}$



$$B_c = \frac{\mu_0 I}{4R}$$


$$B_c = \frac{\mu_0 I}{8R}$$

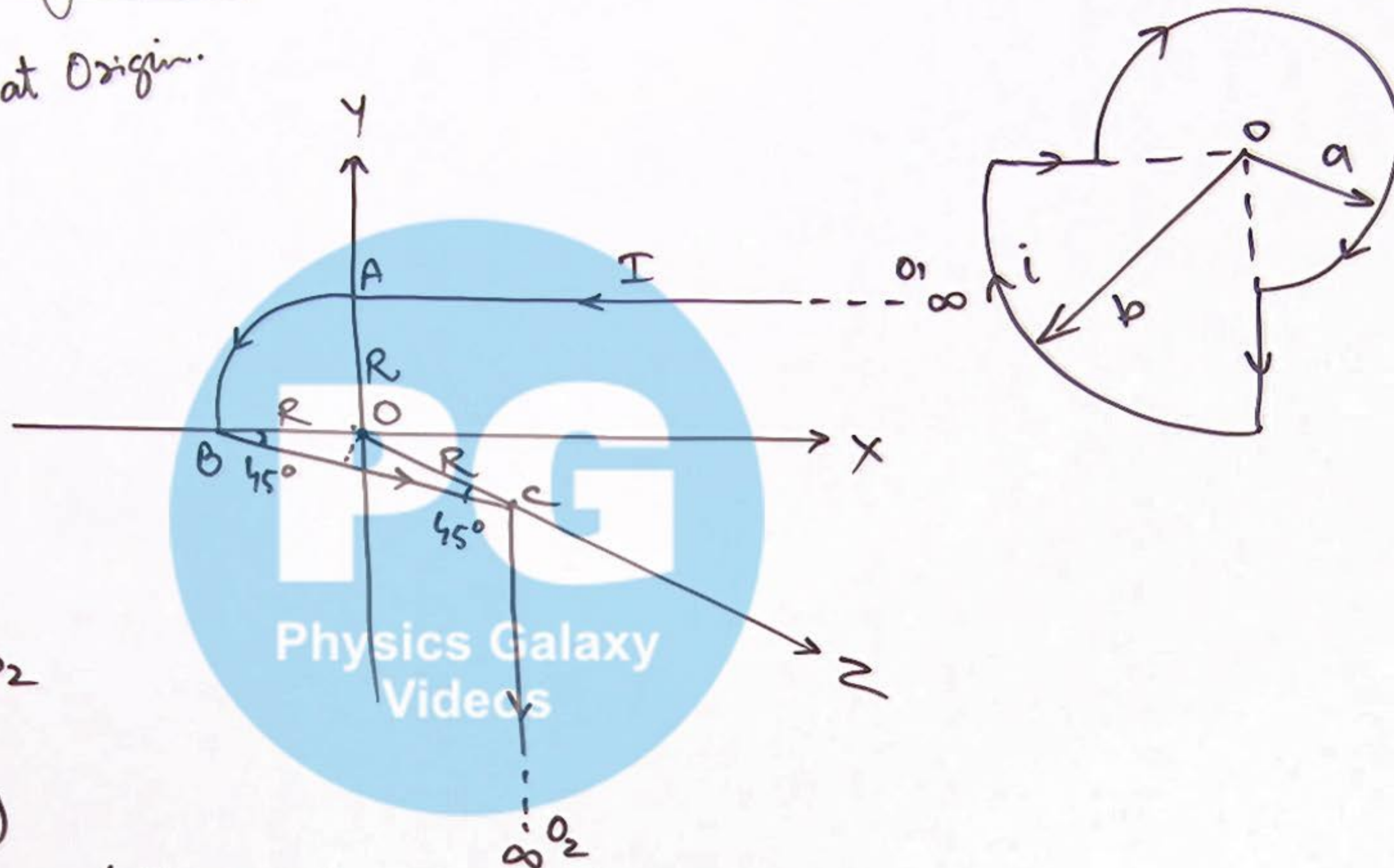
⑧ MF due to 3D wire Segments:

IL: Find MF at Origin.

$$\vec{B}_0 = \vec{B}_{OA} + \vec{B}_{AB} +$$

$$\vec{B}_{BC} + \vec{B}_{CO2}$$

$$\vec{B}_0 = \frac{\mu_0 I}{4\pi R} \hat{k} + \frac{\mu_0 I}{8R} \hat{k} + \frac{\mu_0 I}{2\pi(R/\sqrt{2})} [\sqrt{2}] \hat{j} + \frac{\mu_0 I}{4\pi R} \hat{i}$$



⑨ MF due to a Current Carrying Strip:

Diagram illustrating the magnetic field (MF) due to a current-carrying strip of width b . The strip carries a current I upwards. A point P is located at a distance r from the strip. A differential current element dI is shown at a distance x from the strip. The magnetic field dB is calculated at P due to dI .

Equations shown:

$$dI = \frac{I}{b} \cdot dx$$

MF at P due to dI

$$dB = \frac{\mu_0 dI}{2\pi x}$$

$$B_P = \int dB = \frac{\mu_0 I}{2\pi b} \int_r^{r+b} \frac{dx}{x} = \boxed{\frac{\mu_0 I}{2\pi b} \ln\left(\frac{r+b}{r}\right)}$$

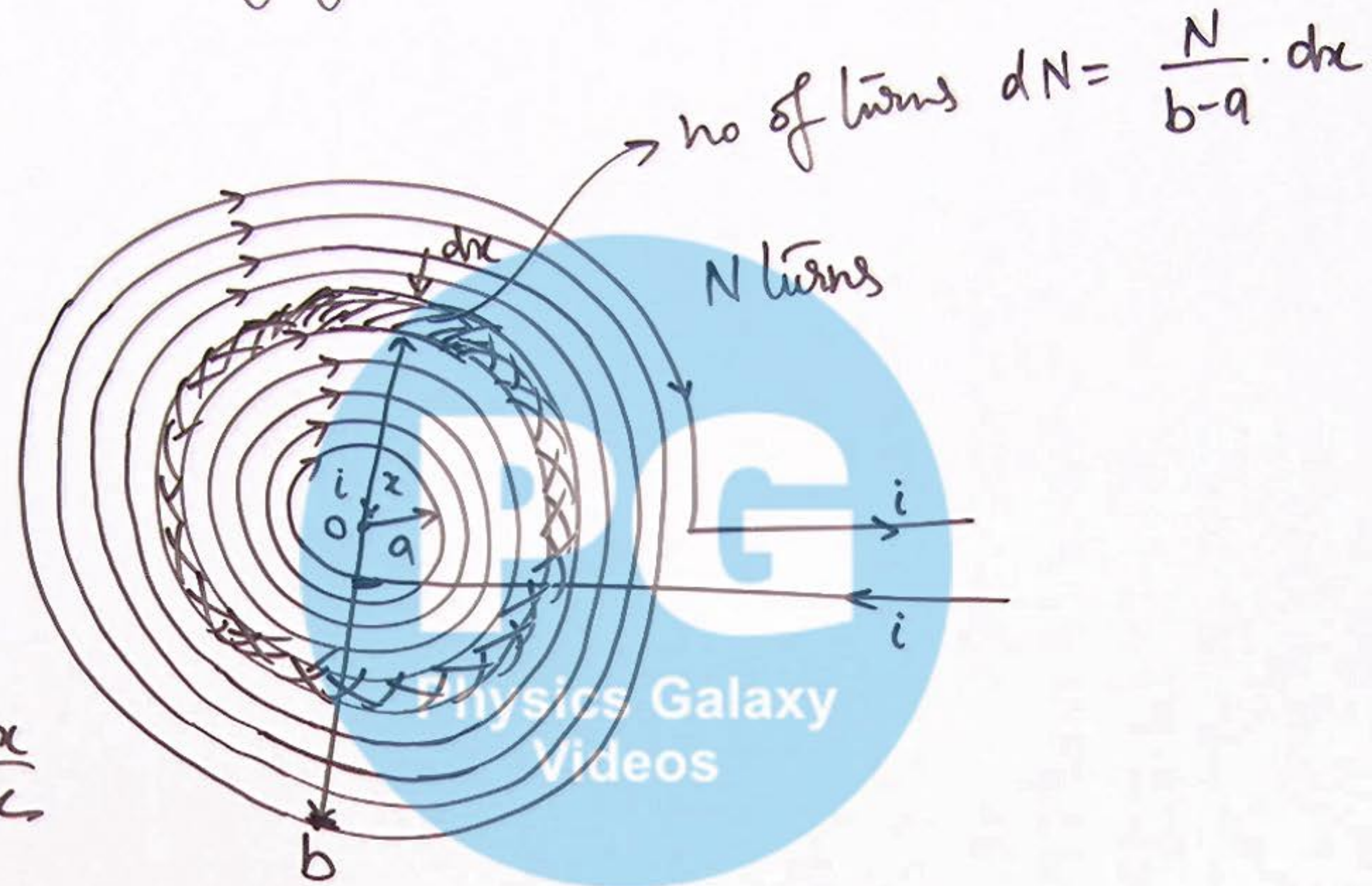
10) MF due to a Current Carrying Spiral:

MF at O due to element dx is

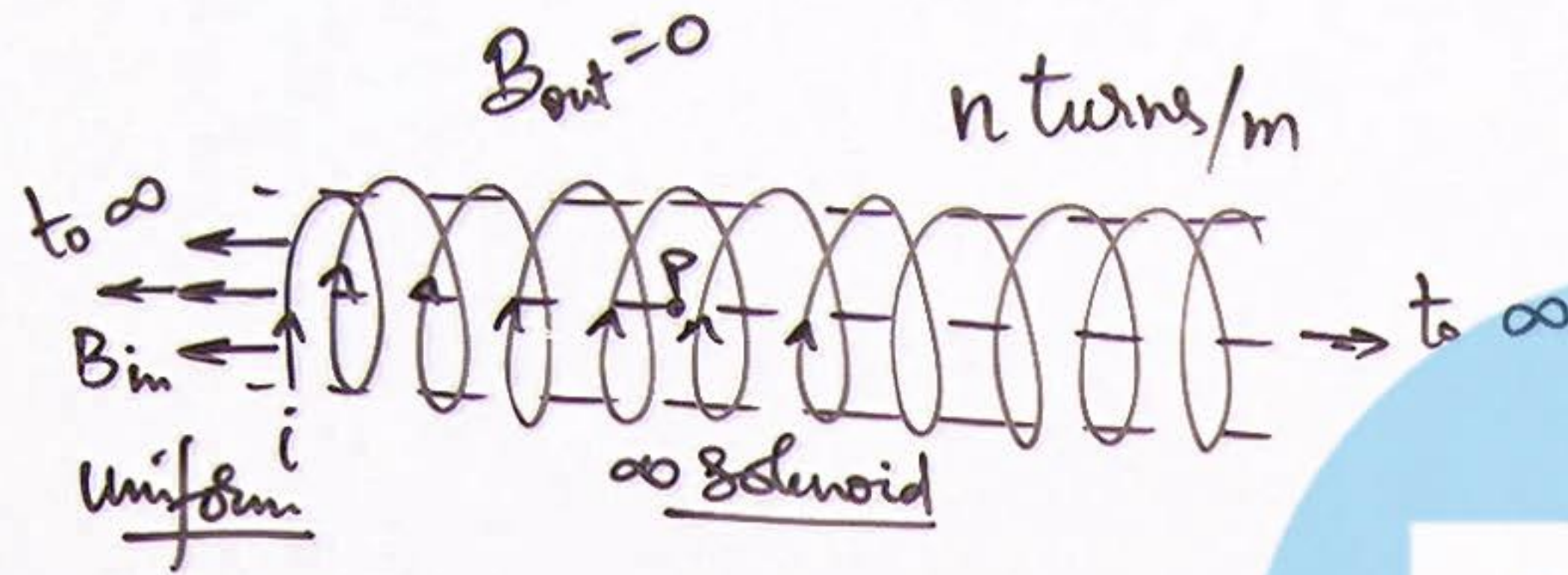
$$dB = \frac{\mu_0 I dN}{2x}$$

$$B_0 = \int dB = \frac{\mu_0 I N}{2\pi(b-a)} \int_a^b \frac{dx}{x}$$

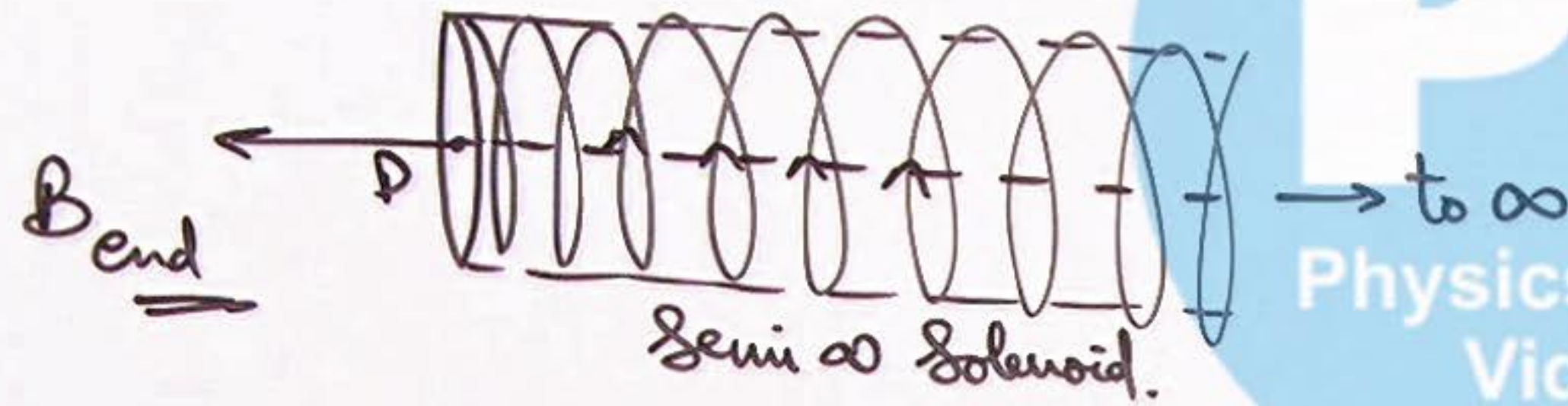
$$B_0 = \frac{\mu_0 I N}{2\pi(b-a)} \ln(b/a)$$



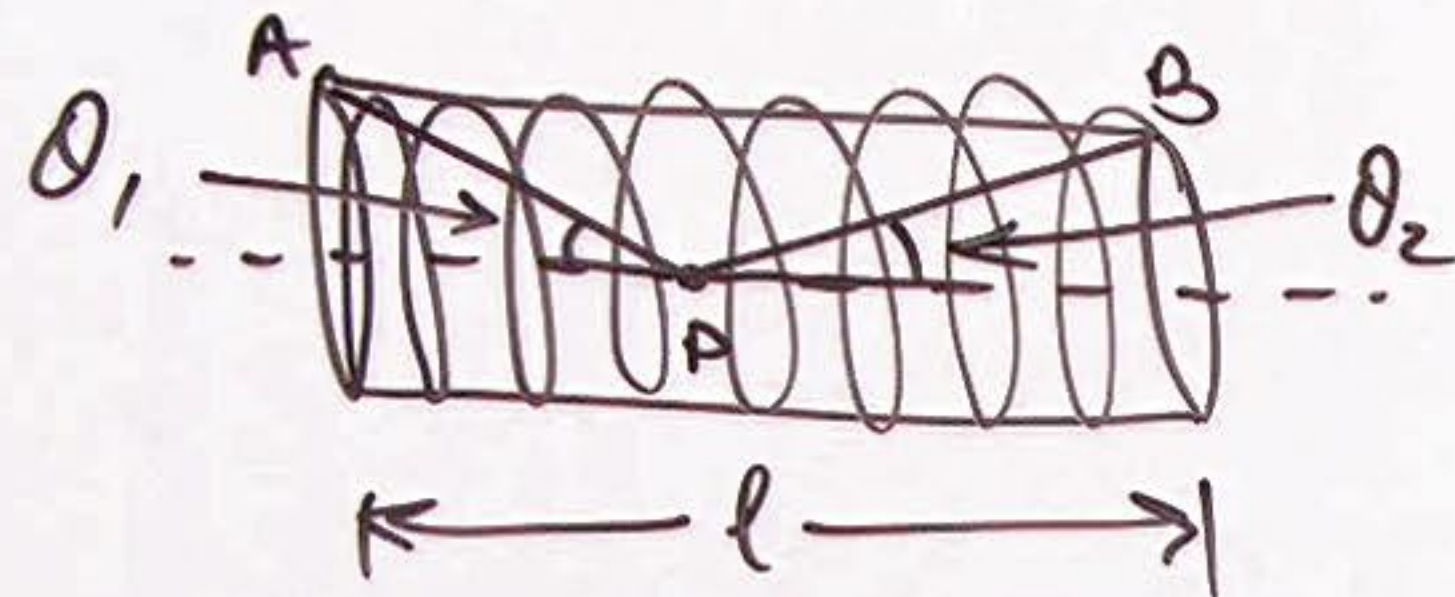
⑪ MF due to a Solenoid:



$$B_{in} = \mu_0 n I$$

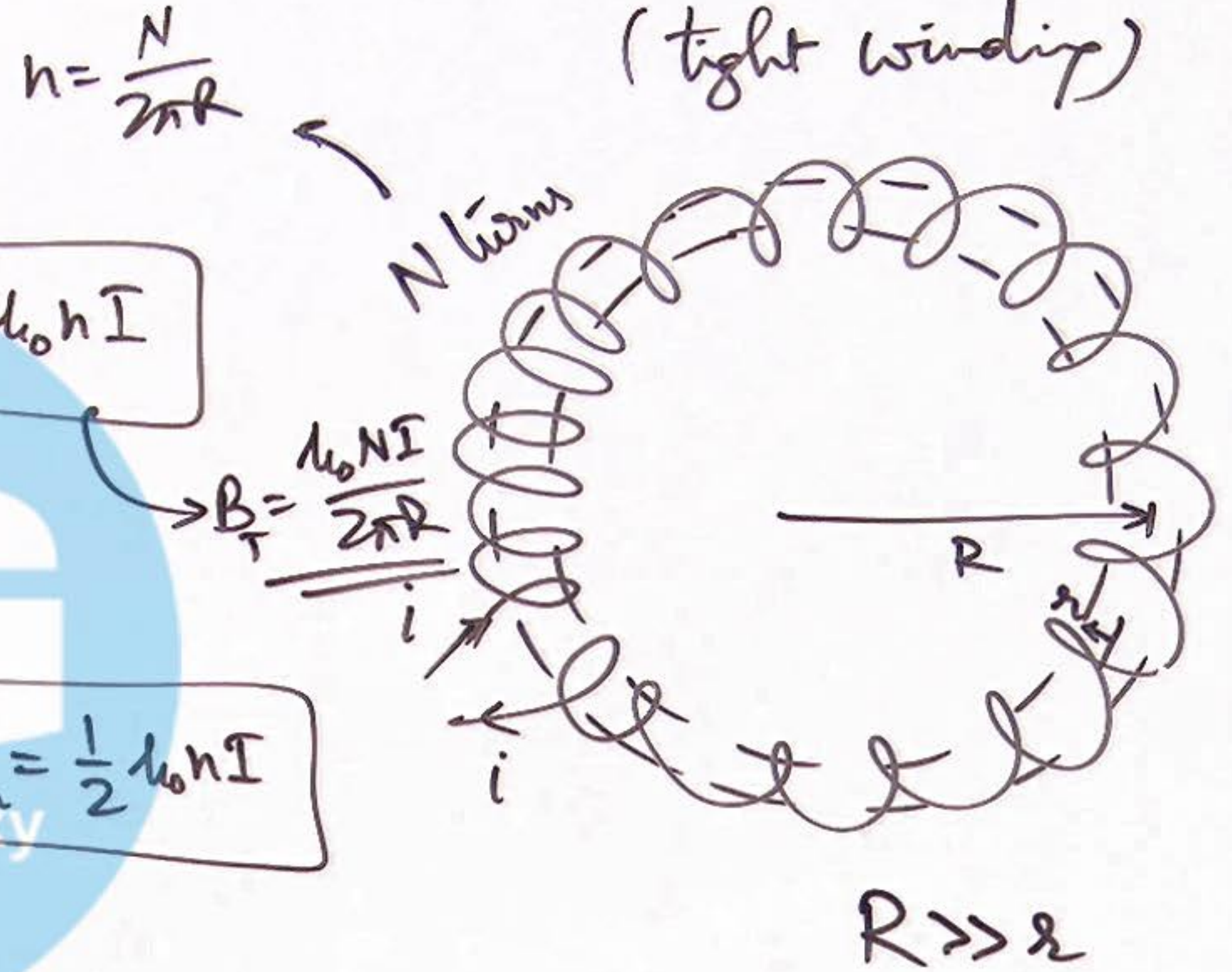


$$B_{end} = \frac{1}{2} \mu_0 n I$$



$$B_p = \frac{1}{2} \mu_0 n I [\cos \theta_1 + \cos \theta_2]$$

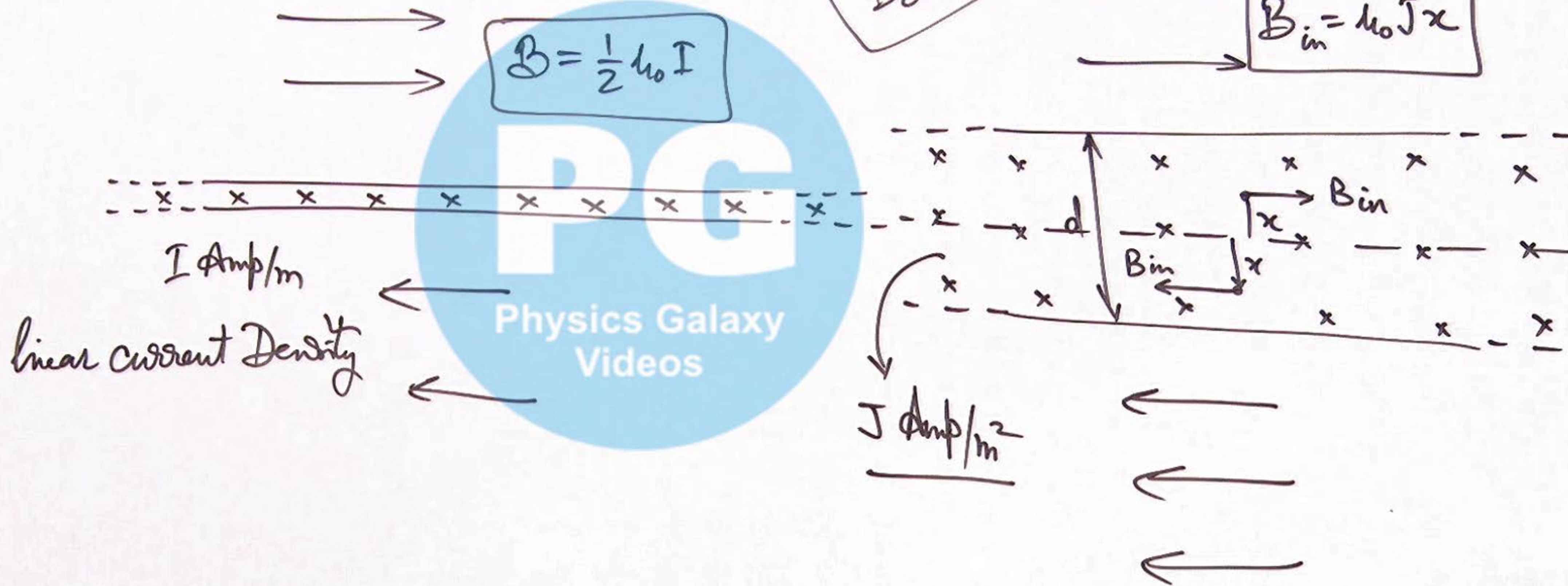
⑫ MF due to a toroid (tight winding):



$$n = \frac{N}{2\pi R}$$

$$B_T = \frac{\mu_0 N I}{2\pi R}$$

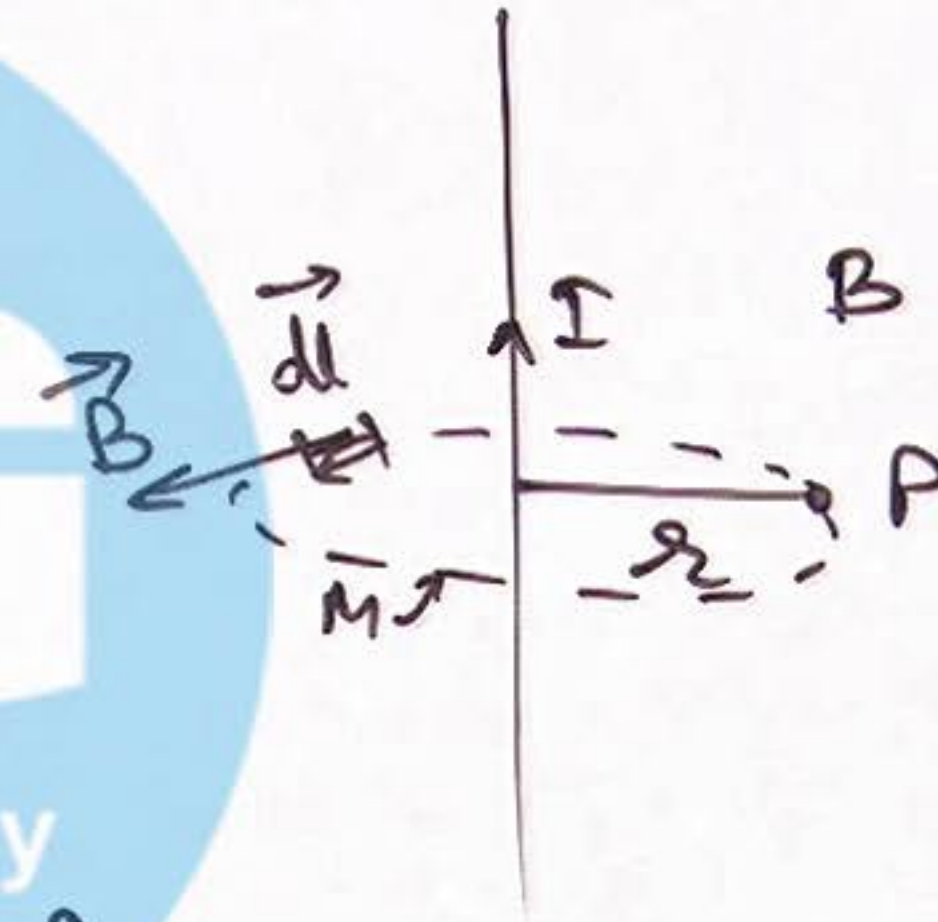
13 MF due to a Current Carrying large sheet :



14 Ampere's Law:

for a closed path

$$\oint_M \vec{B} \cdot d\vec{l} = \mu_0 \sum_M I_{\text{enclosed}}$$



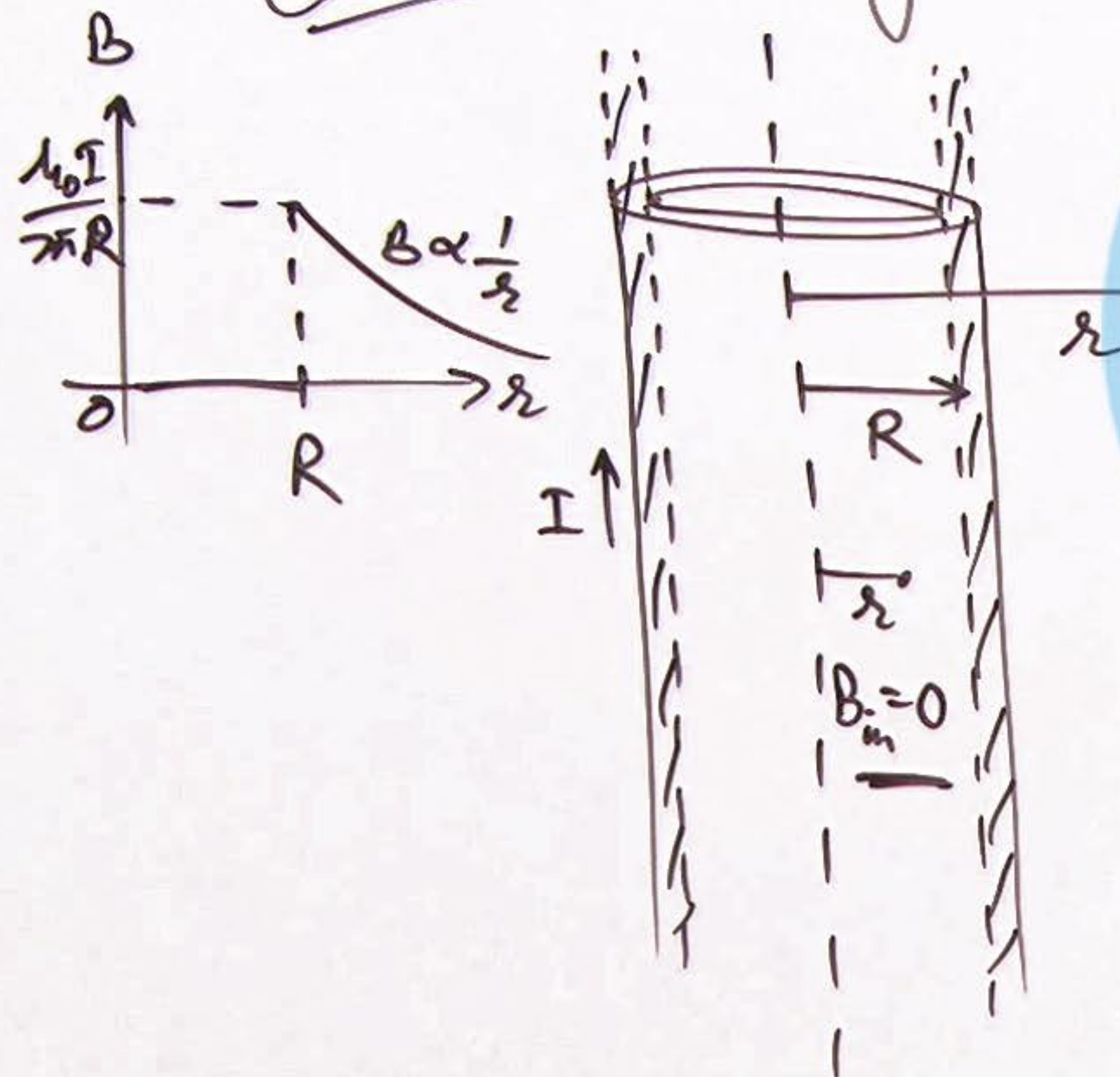
$$\oint_M B dl = \mu_0 I$$

$$B \cdot 2\pi r = \mu_0 I$$

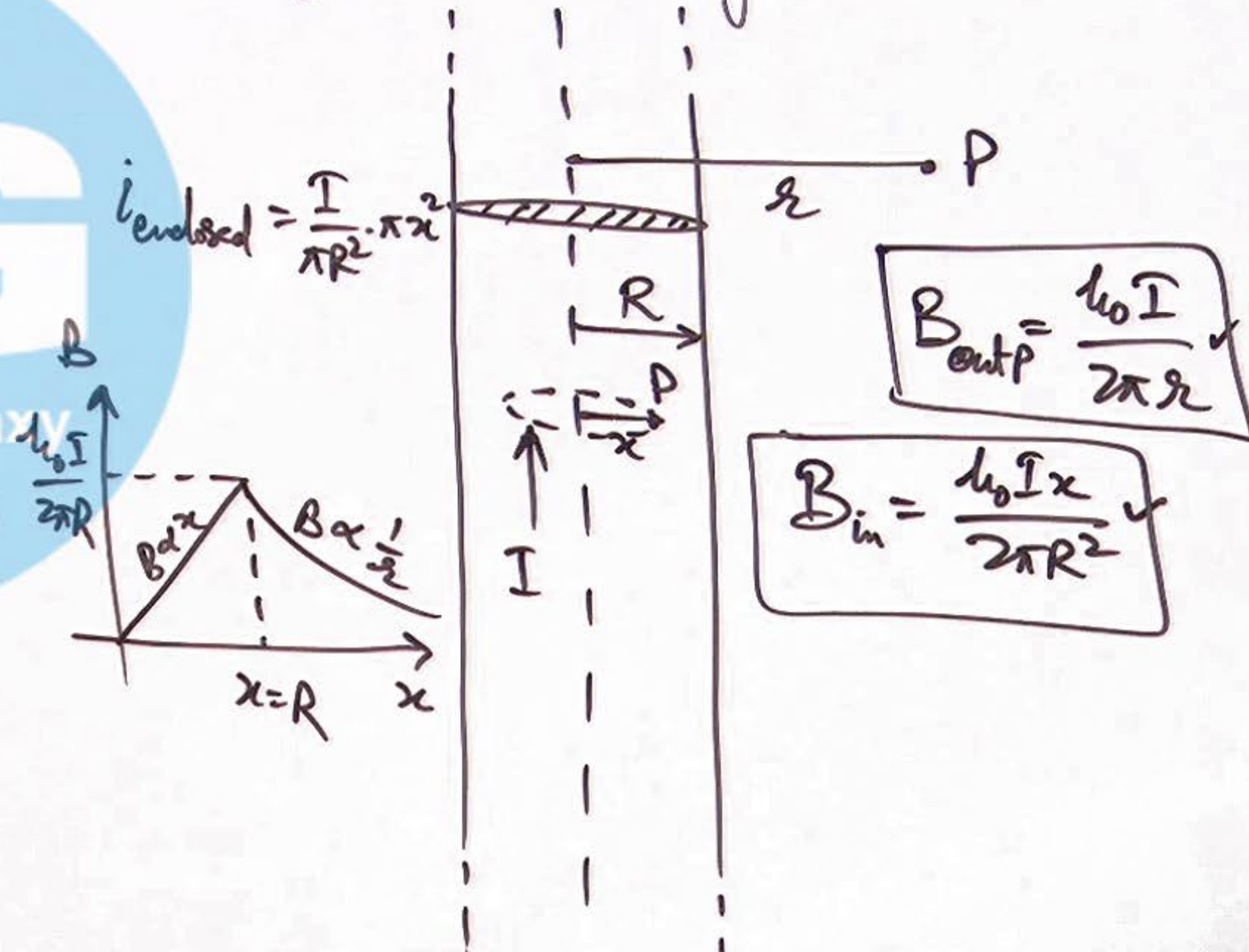
$$\left[B = \frac{\mu_0 I}{2\pi r} \right]$$

(15) MF due to a current Carrying Cylinder:

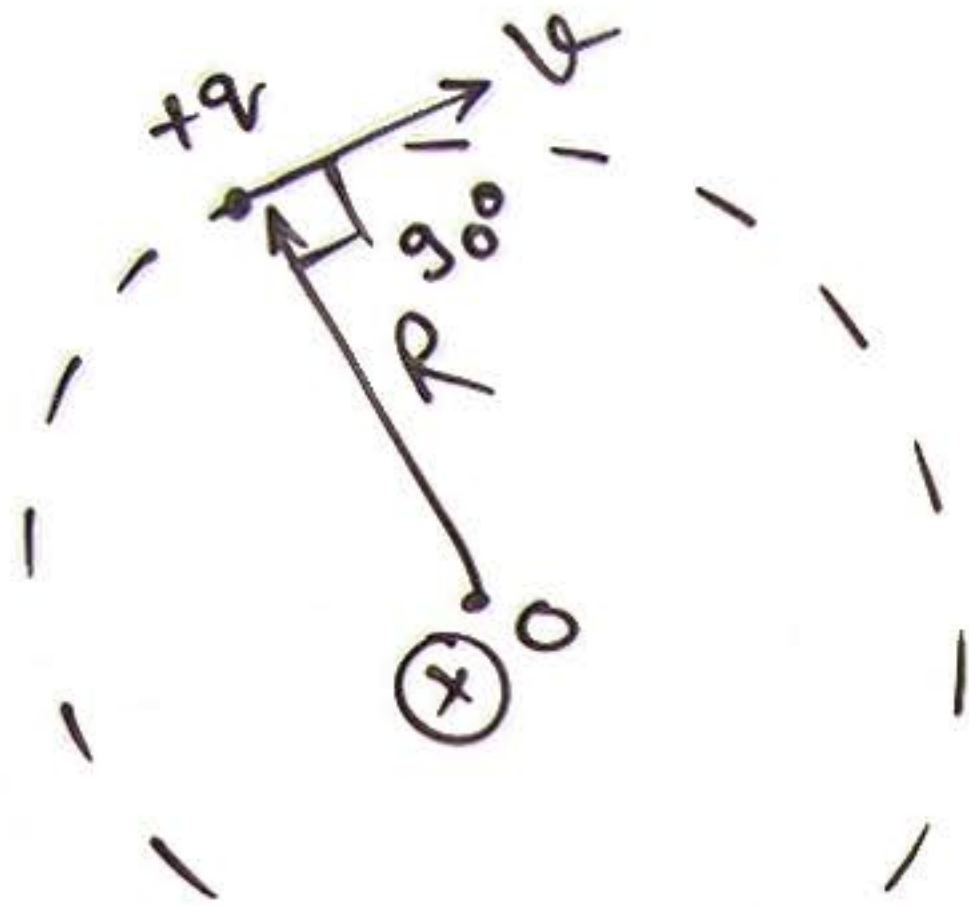
Case-I: Hollow Cylinder



Case-II: Solid Cylinder



16 MF due to a revolving charge:



Equivalent current ^{in orbit} due to Rev of charge

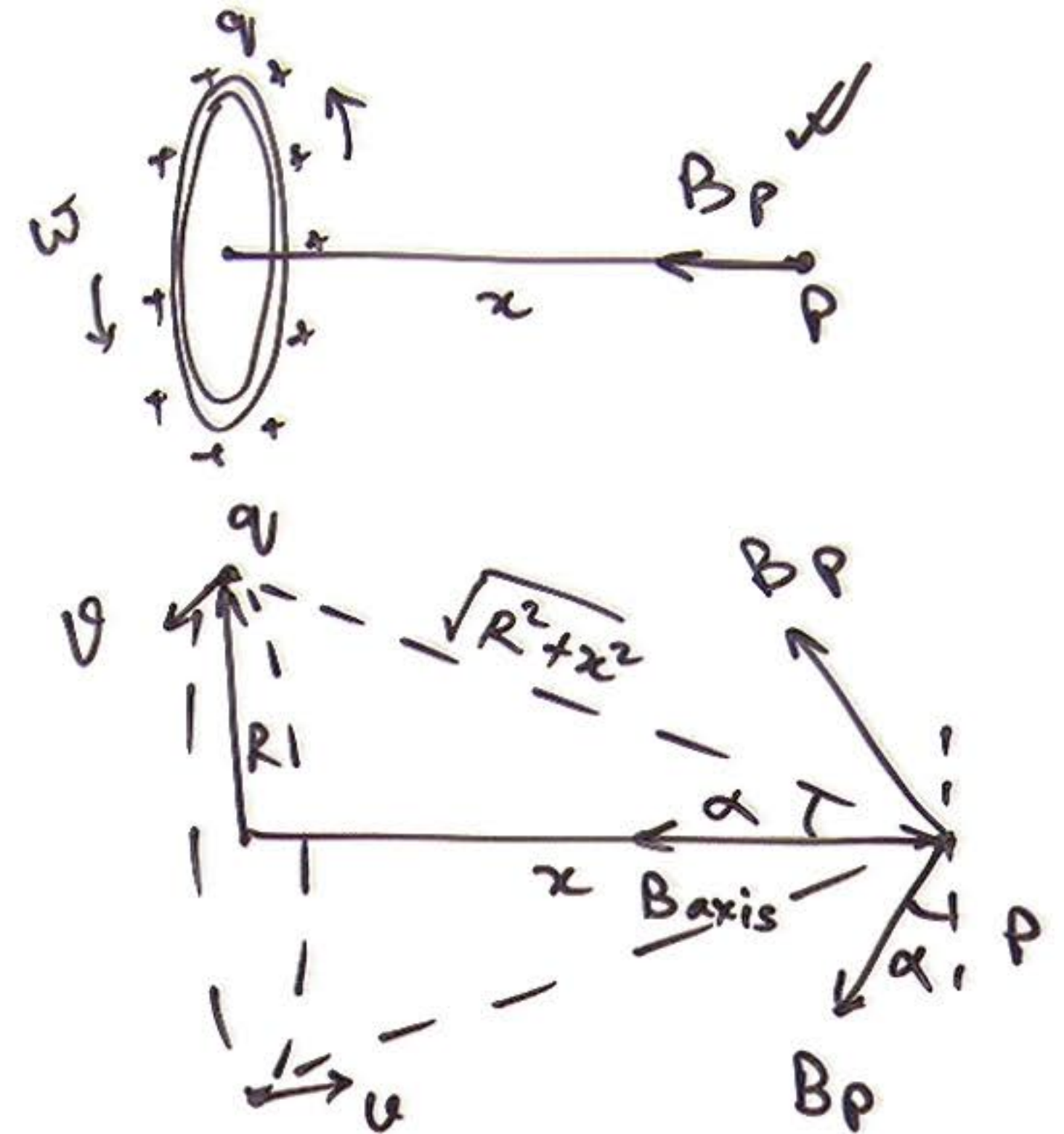
$$B_{\text{axis}} = \frac{\mu_0 I R^2}{2(x^2 + R^2)^{3/2}} = \frac{\mu_0 q v R^2}{4\pi R (R^2 + x^2)^{3/2}}$$

$$B_0 = \frac{\mu_0 I}{2R} = \frac{\mu_0 q v}{4\pi R^2}$$

$$B_0 = \frac{\mu_0}{4\pi} \cdot \frac{q v}{R^2}$$

$$I = \text{charge} \times \text{rev. freq.}$$

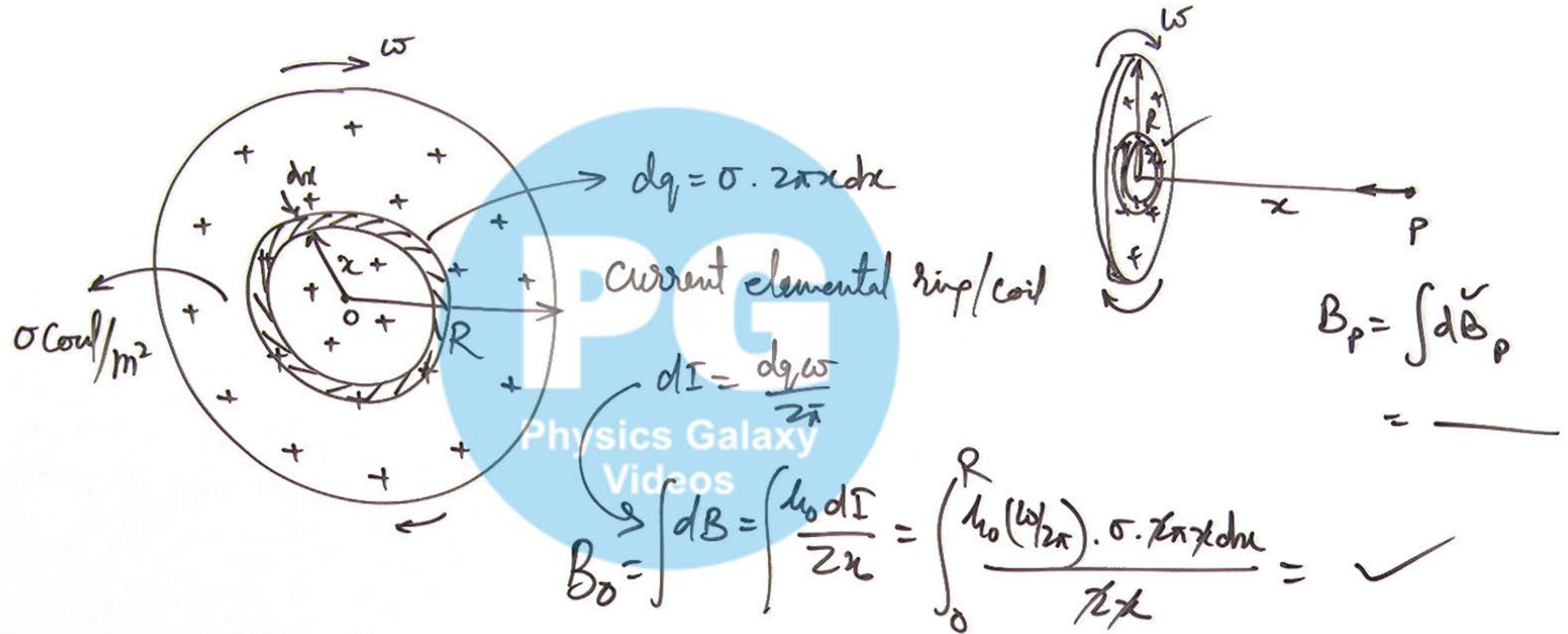
$$I = \frac{q \omega}{2\pi} = \frac{q v}{2\pi R}$$



$$B_P = \frac{\mu_0}{4\pi} \cdot \frac{q v}{(R^2 + x^2)}$$

$$B_{\text{axis}} = \frac{\mu_0 q v R}{4\pi R (R^2 + x^2)^{3/2}}$$

17) MF due to a charged Rotating Disc:



The diagram illustrates the calculation of the magnetic field due to a charged rotating disc. It consists of two main parts: a top-down view of the disc and a side view of an elemental ring.

Top-down view: A circular disc of radius R is shown with a uniform surface charge density σ (Coul/m²). The disc is rotating with angular velocity ω . A concentric circle of radius x is highlighted as a "Current elemental ring/coil". The charge on this ring is $dq = \sigma \cdot 2\pi x dx$. The current in the ring is $dI = \frac{dq\omega}{2\pi}$.

Side view: The elemental ring is shown as a circular loop of radius x with current dI . A point P is located at a distance x from the center of the ring along its axis. The magnetic field at P is denoted by $d\vec{B}_P$.

Equations:

$$B_0 = \int dB = \int \frac{\mu_0 dI}{2x} = \int_0^R \frac{\mu_0 (\omega/2\pi) \cdot \sigma \cdot 2\pi x dx}{2x} = \checkmark$$

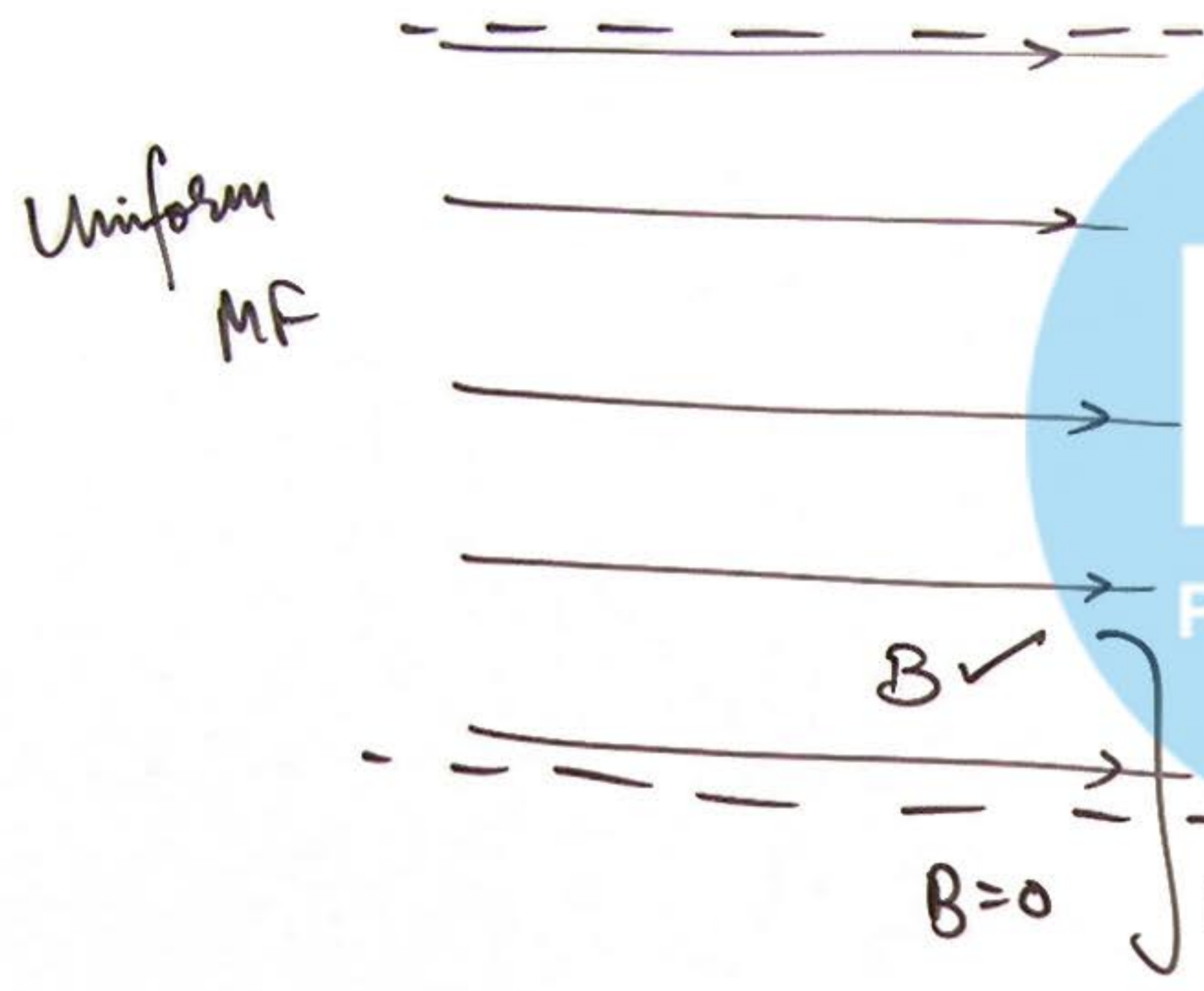
The final result is indicated by a checkmark, suggesting the integration is complete.

Additional notes:

- The magnetic field at point P is given by $B_P = \int d\vec{B}_P$.
- A watermark "PG Physics Galaxy Videos" is visible in the background.

⑮ Variation in Unidirection MF:

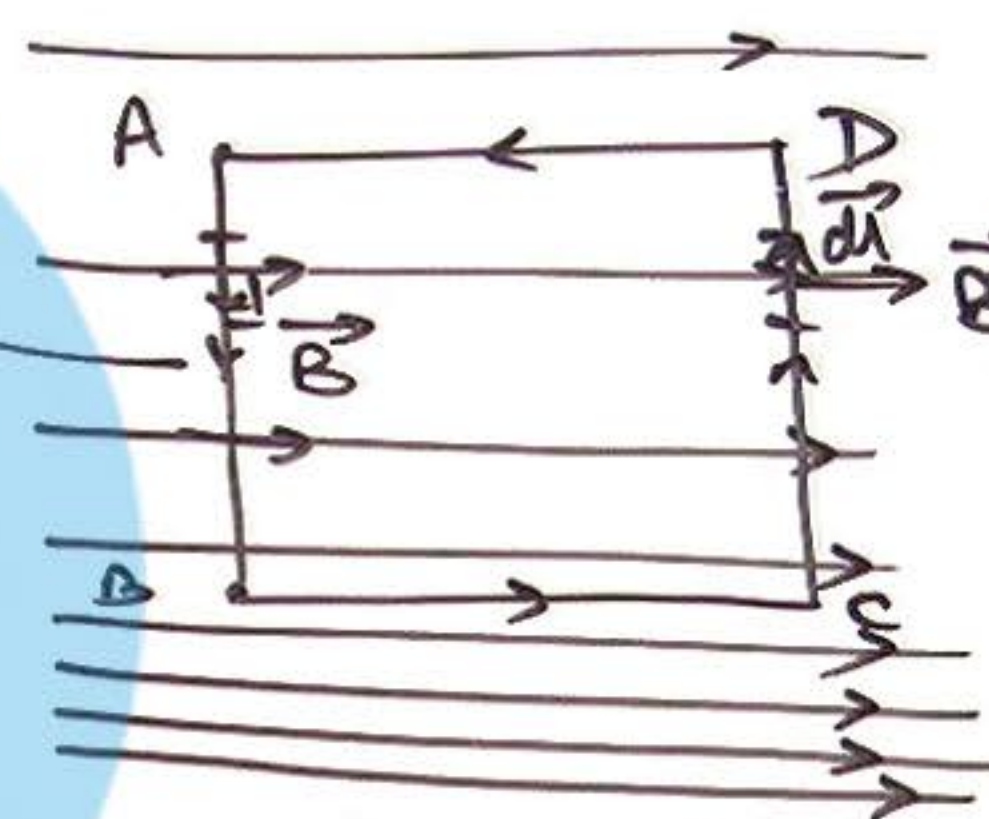
Not possible.



Ampere's law.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

ABCD



$$\int_C \vec{B}_B \cdot d\vec{l} + \int_D \vec{B}_A \cdot d\vec{l} = 0$$

Not possible!

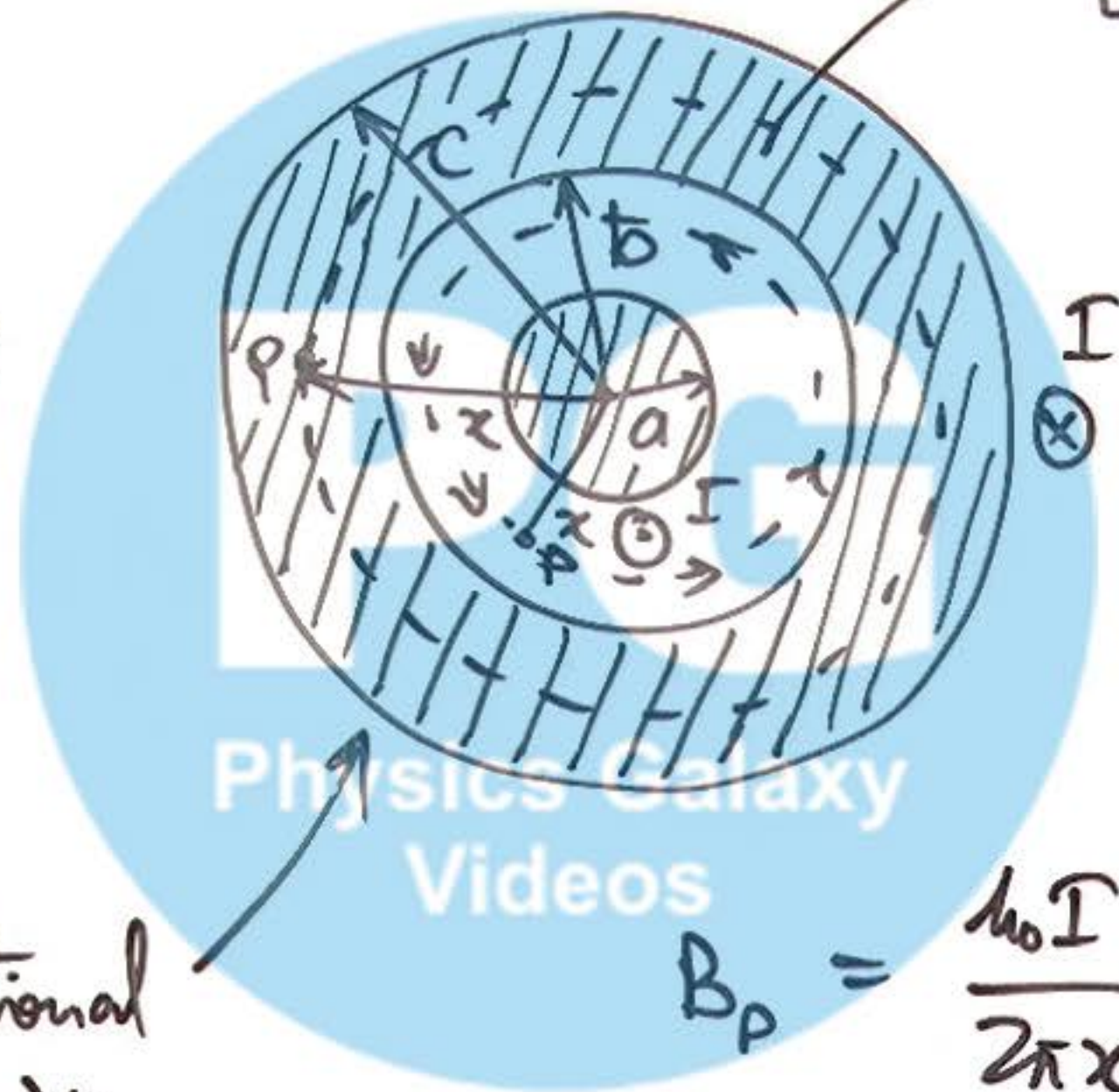
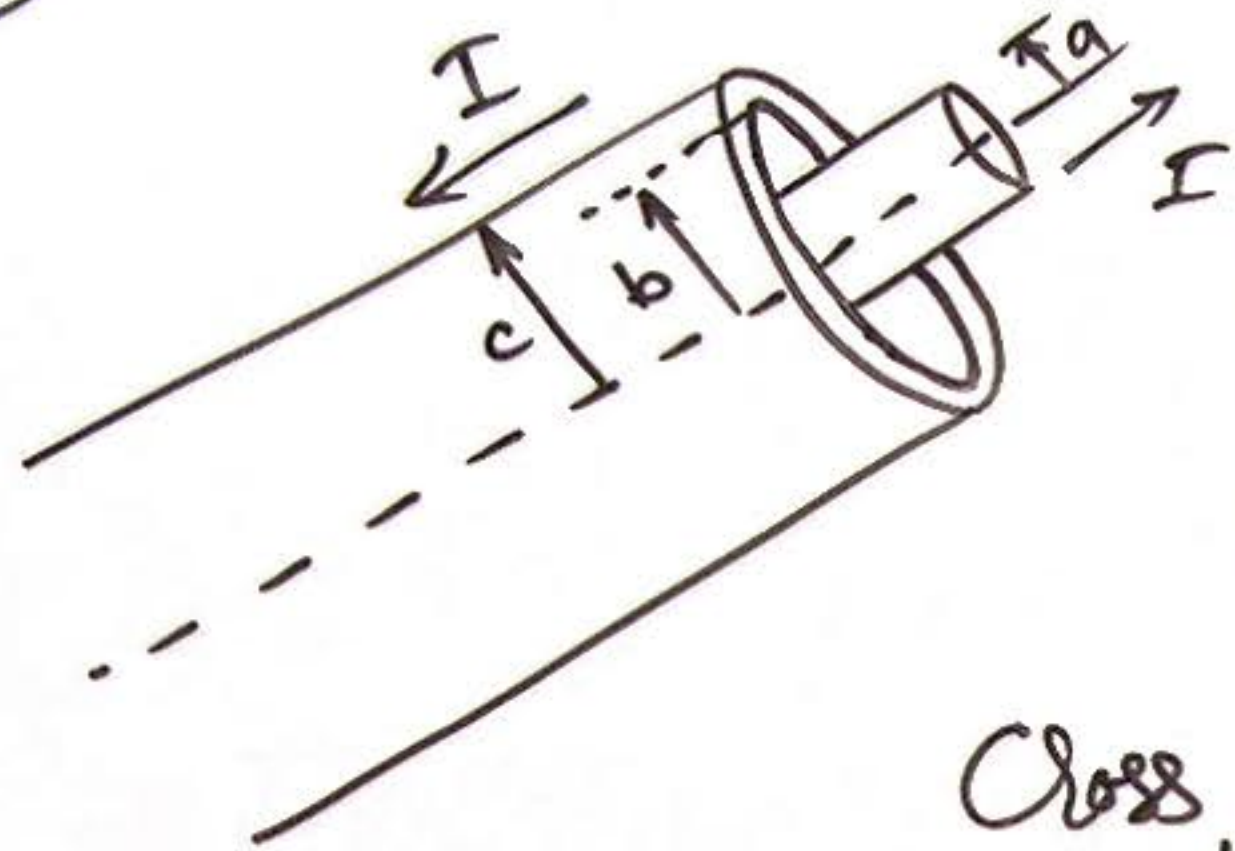
$$\int_A \vec{B} \cdot d\vec{l} = \int_C \vec{B} \cdot d\vec{l} = 0$$

because $\vec{B}$ is $\perp$ to $d\vec{l}$

Physics Galaxy Videos

$$B_A < B_B$$

19) MF due to a Coaxial Cable:



$$I_{\text{enc}} = I - \frac{I}{\pi(c^2 - b^2)} \times \pi(x^2 - b^2)$$

$$B_{\text{out}} = 0$$

20) MF inside a Cavity of a Cylindrical current Carrying Conductor:

Cross
Sectional
View

$$B_p = \frac{\mu_0 I}{2\pi x} \quad \underline{a < x < b}$$

$$B_{\text{in}} = \frac{\mu_0 I x}{2\pi a^2} \quad x < a$$

$$B_p = \frac{\mu_0 I_{\text{enc}}}{2\pi x} \quad b < x < c$$